

XGBoost based solutions for detecting fraudulent credit card transactions

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Abstract—Due to the emigration of businesses to the internet, credit cards have become a widely used tool for customers to pay for purchases both online and offline. However, fraudsters try to attack these payment tools using various techniques, and credit card fraud transactions have become dangerous. Credit card payment security is becoming an important research topic, and different ways are proposed to overcome this problem. In this article, we will propose a solution that combines the strengths of three methods to secure credit card fraud transactions. The Differential Evolution (DE) algorithm is used to select the best hyperparameters of the XGBoost algorithm (eXtreme Gradient Boosting). Synthetic Minority Oversampling (SMOTE) combined with Edited Nearest Neighbour (ENN) to handle the imbalanced data issues. Finally, we will classify fraud transactions using the optimized XGBoost algorithm. The proposed solution is evaluated using the European data set in terms of Accuracy, Recall, F1-score, and AUC. As a result, the proposed solution generally achieved the best score in terms of all the measurements used in this study. In terms of accuracy, we achieved 99.94%, similarly, we reached the best score in terms of the Precision score, which is 80.68%, in addition, we got 86.02% of recall score, 83.27 of F-measure, and 99.21 of AUC score. Thus, we emphasize the superiority of our proposed method over the competitive machine learning model utilized in this work.

Index Terms—Fraud Transactions Detection, Hyperparameters Optimization, Differential Evolution Algorithm, XGBoost algorithm, Synthetic Minority Oversampling, Edited Nearest Neighbor

I. INTRODUCTION

Studies in credit card fraud detection has become an important research field in the world. Banks and financial institutions spend a lot of money researching efficient and powerful methods to make the payment process that uses credit cards more secure for clients. Likewise, researchers have developed various techniques and methods to stop the crime committed by fraudsters using various algorithms. Currently, machine learning algorithms have become more powerful and effective tools for solving more complex problems in our world, due to the huge amount of data available from the enterprise and the advancement of hardware capability. Today, one of the most complex problems is credit card fraud transactions [1].

A credit card is a physical medium used by a bank or credit union to allow cardholders to pay a merchant for goods and services online [2]. Financial institutions and banks have suffered from a number of fraud transactions committed every year [3]. Credit card fraud occurs when card identities, such as credit card number, expiration date, password, and credit card holder name, are stolen by fraudsters to make fraudulent transactions [4]. Machine learning and deep learning techniques show their efficiency in distinguishing between normal

and abnormal transactions [5], [6]. There are two methods to detect fraudulent transactions, the first method uses supervised learning algorithms [7], such as neural networks [8], regression model [9], naive bayes [10], decision trees [11], differential evolution [12], fuzzy logic [13], genetic algorithm [14], and particle swarm optimization [15]. The second method uses unsupervised algorithms [16], such as self-organizing maps (SOMs) [17], K-Means [18].

An important thing in the process of detecting fraudulent transactions is the problem of imbalanced data sets [19]. Many techniques are used to deal with the problem of unbalanced datasets [20]. These techniques can be classified into two categories, under-sampling methods and over-sampling methods. In the under-sampling method, many majority classes are ignored from the training set to balance the training data, such as the Random Under-Sampling method (RUS) [21]. The over-sampling method generates more samples of the minority classes to balance the training set, such as SMOTE. In this article, we will combine the two categories using SMOTE and ENN to get more accurate results and to overcome the imbalanced data sets issue [22].

In this work, we will propose a solution for fraud transaction detection in credit card data sets. The key contributions of this paper are the following:

- First, we will use SMOTE-ENN to overcome the imbalanced credit card data sets issue.
- Second, we will select the best hyperparameters for the XGBoost algorithm using the DE algorithm based on the cross-validation technique.
- Finally, we will classify the testing set using the optimized XGBoost.

The remainder of this paper is organized as follows. Section presents some related work II. Section III presents the methodology and approach adopted. The proposed solution is described in subsection III-C. The experiment results are discussed in Section IV. Finally, Section V includes concluding remarks.

II. RELATED WORK

Credit card fraud detection has earned a lot of attraction, and there are many published papers regarding credit card fraud detection. Recent studies use machine learning and deep learning techniques and different optimization techniques to achieve state-of-the-art results for credit card fraud detection. In this section, we present some related work in this field and especially the studies that used credit card data sets. For example, Rtayli *et al.* [23] proposed a hybrid approach

that combined various techniques to detect fraud transactions. Recursive feature elimination is used to select the predictive features using support vector machine learning algorithms, Gridsearchcv to optimize the hyperparameter optimization, and SMOTE to overcome the imbalanced data set problem. The results obtained on many real-world data sets showed that the proposed solution is efficient in classifying fraud transactions compared to the state-of-the-art approaches. Li *et al.* [24] developed an approach to deal with the issue of class imbalance based on the divide-and-conquer algorithm. They trained a model of anomaly detection in the minority class to eliminate both a few outliers of the minority samples and many majority classes from the original dataset. The remaining class form an overlapping subset that has a low imbalance ratio and a lower learning interference from both the minority class and the majority class compared to the original dataset. The obtained results showed that the developed approach outperforms state-of-the-art ones based on the Dynamic Weighted Entropy (DWE) measure.

Askari *et al.* [25] proposed a novel model for the detection of online fraudulent transactions. They developed an IFDTC4.5 decision tree using intuitionistic fuzzy logic and a C4.5 decision tree classifier. The results obtained showed that the detection rate is great and powerful compared to those in recent existing works. In [26], Jiang *et al.* proposed a novel method to detect fraud transactions consisting of four stages. First, they divided all cardholders into different groups based on historical online transaction data. After that, they used a window sliding strategy to aggregate the online transactions in each group. Next, they extract a collection of specific behavioral patterns for each cardholder based on the aggregated online transactions and the cardholder's historical online transactions. Finally, various classifiers are trained in each group, and it is used to detect online fraud transactions. If a new online transaction is fraudulent, a feedback mechanism is taken in the detection process to solve the problem of concept drift. The conducted experiments results showed that the proposed method performs better in comparison with others methods.

Sanober *et al.* [27] combined spark with a deep learning model as a solution for fraud transaction detection. In addition, they used various machine learning models such as random forest, support vector machine, decision tree, logistic regression, and k-nearest neighbor for comparison. For evaluation, they used the European credit card data set and the obtained results achieved an accuracy of 96%. In [28], Chen *et al.* presented a novel technique for discriminating between fraud and legitimate transactions. The idea of this technique aims at using a deep-convolution neural networks (DCNN), after that, the proposed solution is compared with an auto-encoder model and other deep learning models using real-time credit card data set. In general, the results obtained prove that the proposed solution outperforms the other machine learning models. In contrast to prior work, our solution uses hybrid methods to optimize and enhance credit card fraud transactions. The results obtained show the strong ability of our solution in detecting fraud transactions in credit card transactions.

III. METHODOLOGY AND APPROACH

A. Data Description

The data set used to evaluate our solution and test its performance contains transactions made by European cardholders for two days in September 2013 [29]. This data set is collected by two researchers, as part of a partnership between World Line Company, a specialized money transfer service provider, and Liber Brussels University. The number of transactions in this data set is 284315 transactions. The number of features is 31 features, which are time, amount, class, $V_1, \dots, V_{23}$.

The target variable of this data set takes two values 0 if the transaction is legitimate and 1 otherwise.

B. Data preprocessing

In this work, our proposed solution has been evaluated using a data set that contains credit card transactions. Data preprocessing is the first step in every machine learning project pipeline, and the most consuming part, their objective is to make the data understandable by the algorithm. The used data set consists of 31 features and 284315 samples. All the features have been scaled except for Time and Amount features; we are using Robustscaler methods [30] to scale them. In addition, this data set has no missing value and all features are numerical. After that, we split the data into two subsets: the training set which contains 70 percent of transactions, and the testing set 30 percent of transactions. As we can see from figure 1 this data set is imbalanced, to overcome this issue we are utilizing SMOTEENN oversampling methods.

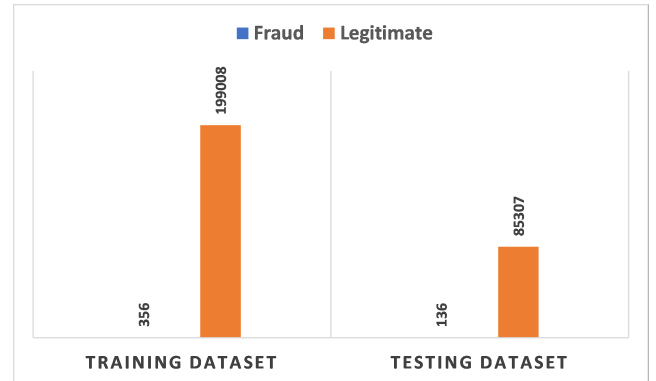


Fig. 1. Distribution of transactions per target variable

C. Proposed solution

Through this section, we present the two stages of our solution used to detect fraudulent transactions: Differential Evolution algorithm for hyper-parameters tuning for selecting the best hyper-parameters, Syntactic Minority Oversampling combined with Edited Nearest Neighbor (SMOTEENN) is used to balance the training set. Figure 2 illustrates the proposed system diagram and the details for each step is given in each subsection.

1) *Data sampling*: A highly imbalanced data set, which occurs in many real-world applications, especially in credit card fraud transactions data set, makes machine learning algorithms classification difficult. The over-sampling and under-sampling approaches help to handle this issue. In this paper to overcome the imbalanced issue in our data set, we have used the over-sampling technique SMOTE and the under-sampling method ENN.

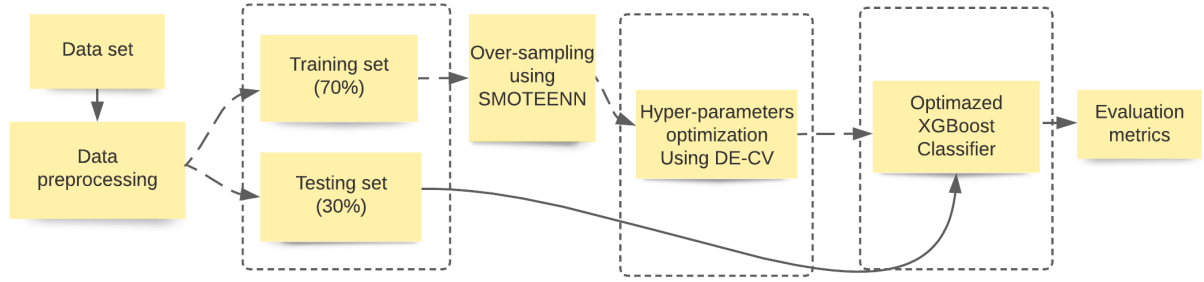


Fig. 2. Flow diagram of our proposed solution

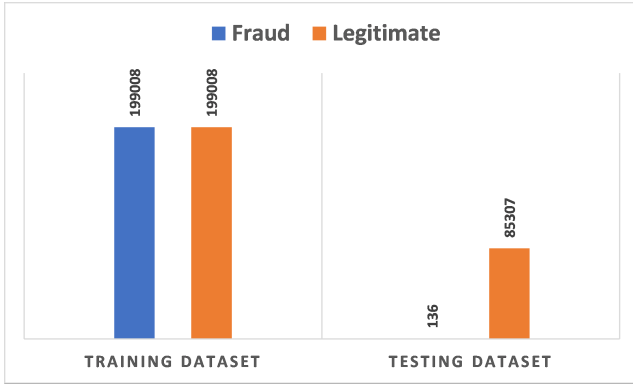


Fig. 3. Distribution of transactions per target variable after applying SMOTEENN method

Figure 3 shows the distribution of our training dataset after applying the SMOTEENN technique. From this figure, we can see clearly that the two classes have the same number of fraud and legitimate transactions in the training dataset. So our training dataset is balanced and ready for the training task. The SMOTEENN generated 198 652 fraudulent transactions based on the 356 real fraud transactions in the training subset. Those new fraud transactions generated are added to the training set.

2) *Differential Evolution Algorithm*: The mechanism of this algorithm is as follows: First, a randomized population is initialized and it is evaluated using the fitness function. Afterward, we update the population and create new candidate solutions by combining existing ones according to their simple formula, and then keeping whichever candidate solution has the best fitness on the optimization problem at hand [31]. The used algorithm is detailed in 1.

TABLE I
OPTIMIZED HYPERPARAMETERS

Name	Range
n_estimators	□ 300, 500 □
max_depth	□ 1, 10 □
learning_rate	□ 0.01, 0.7 □

Table I describes the optimized hyperparameters using differential evolution algorithms based on the crossover validation method. The optimized hyperparameters are n_estimators which denotes the number of Decision Trees in the Xgboost algorithms. The second hyperparameter is the max_depth which refers to the longest path between the root node and the leaf node in each decision tree. The third hyperparameter

Algorithm 1 Pseudocode of DE.

Input: Fitness function (f), $lower_bounds$, $upper_bounds$, N_p , max_iter , F , P_c

A random population P is Initialized

Evaluate fitness f of P

for $t = 0, \dots, max_iter$ **do**

for $i = 0, \dots, N_p$ **do**

 Mutation operation expands the search space. A mutant solution vector donor D_i of a chromosome X_i is produced by

$$D_i = X_{r_1} + F(X_{r_2} + X_{r_3})$$

Where $r_1, r_2, r_3 | r_1 \neq r_2 \neq r_3 \neq i \in \{0, 1, \dots, N_p\}$ are randomly choosing solutions and $F \in [0, 2]$ is a scaling factor and i is the index of current solution.

In crossover phase the parent vector are mixed with the mutated vector to produce a trial vector by the formula below

$$U_i = \begin{cases} D_i & \text{if } rand(0, 1) \leq P_c \text{ or } i = \delta \\ X_i & \text{if } rand(0, 1) > P_c \text{ and } i \neq \delta \end{cases}$$

P_c is the crossover probability, δ is randomly selected variable location.

end for

for $i = 0, \dots, N_p$ **do**

if $U_i \notin (lower_bounds, upper_bounds)$ **then**

$U_i = \max(\min(U_i, upper_bounds), lower_bounds)$

end if

 Evaluate the fitness $f(U_i)$ of U_i

 Perform greedy selection using $f(U_i)$ and f_i to update P by the below formula

$$\text{if } f(U_i) < f_i, \text{ then } \begin{cases} X_i = U_i \\ f_i = f(U_i) \end{cases}$$

end for

end for

is the learning_rate which is the shrinkage you do at every step you are making; their goal is to control the weighting of new trees added to the model.

3) *Model Evaluation*: In this paper, we are using several evaluation metrics to evaluate our proposed solution, assess its performance, and compare it with the performance of the state-of-the-art models. These measurement metrics are described in Table II as follows;

Where tp (true negatives) denotes the number of correctly classified normal transactions, fp (true positives) refers to the

TABLE II
EVALUATION METRICS FORMULAS

Name	Formula
Accuracy	$\frac{tp + tn}{tp + fp + tn + fn}$
Precision	$\frac{tp}{tp + fp}$
Recall	$\frac{tp}{tp + fn}$
F-Measure	$\frac{2 \times Precision \times Recall}{Precision + Recall}$

number of normal transactions incorrectly classified as abnormal transactions, fn (true negatives) is the number of abnormal transactions incorrectly classified as normal transactions, and tp (true positives) denotes the number of correctly classified normal transactions. Otherwise, AUC is a measurement that is used in a classification problem to measure the performance of a model. In this work we are utilizing this measure to show the ability of our model in separating fraud from non-fraud transactions. The closer the AUC score to 1, the better the performance in distinguishing normal from abnormal transactions.

IV. EXPERIMENTAL RESULTS

To investigate the importance of our solution in detecting fraud transactions in credit cards, we have conducted an experiment using the state-of-the-art machine learning algorithms excited in the literature for comparison. These classifiers are LGBMClassifier (LGBM), AdaBoost (AD), GradientBoosting (GB), RandomForest (RF), Decision Tree (DT), KNeighbors (KNN), LogisticRegression (LR), CatBoostClassifier (CB), GaussianNB (NB).

Table III shows the obtained results for our experiments conducted in which we compare our solution with nine machine learning algorithms. It can be seen that the best accuracy can be achieved using our solution which is 99.94 percent of transactions that are correctly classified, so our model can distinguish between legal and illegal transactions in the credit card dataset. The lowest score is achieved from the LR which is 97.31%. Moreover, the proposed solution got the highest score (80.60 percent) of precision. Thus, we emphasize the strong ability to classify fraud transactions correctly using our model. In addition, the lowest precision score is reached from LR (5.27%) of fraud transactions that are correctly identified. The second heist precision score is 47.28% from the LGBM model. To summarize, all competitiveness is not able to classify more than 50% of fraud transactions correctly. In contrast, our suggested technique can classify 80 percent of them. In terms of Recall, we reached 86%, the heist score is got from the GB model (94.11 percent). Additionally, our model achieved the highest F measure (83.22%) and the highest AUC score (99.21%). In general, from these results, the superiority of our model against the competitive model is highlighted. And prove the efficiency of hyperparameters optimization using differential evolution algorithms for enhancing the performance of the proposed algorithm.

In Figure 4, we have plotted the evaluation metrics of the used state-of-the-art algorithms. From this figure, we can see that the proposed solution performed better results in terms of the used evaluation metrics.

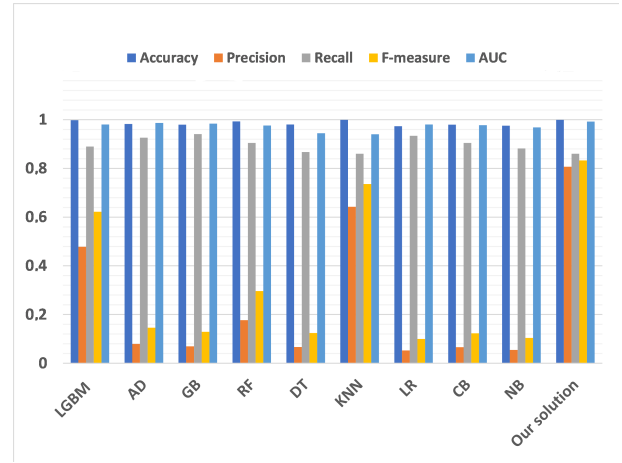


Fig. 4. Comparison with the state-of-arts machine learning

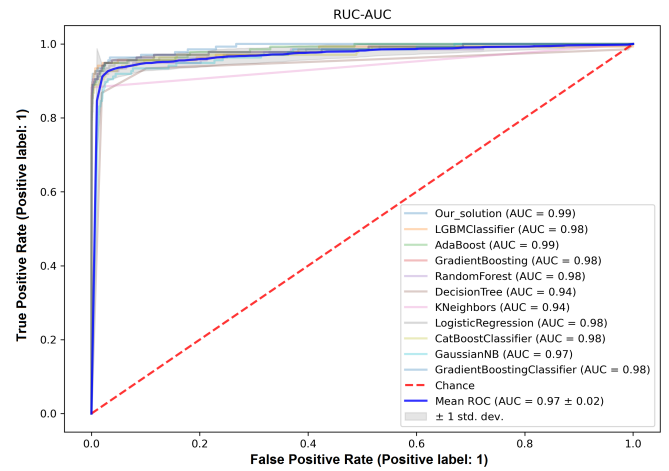


Fig. 5. AUC curve for the proposed solution

Figure 5 illustrates the AUC curve. Overall, it is clear that the proposed solution curve is located in the figures' top-left corner which proves the ability of our proposed solution in detecting fraud transactions correctly. In addition, Figure 6 presented the Precision-Recall curve. From this figure, we can notice the superiority of our solution in comparison with the other classifier.

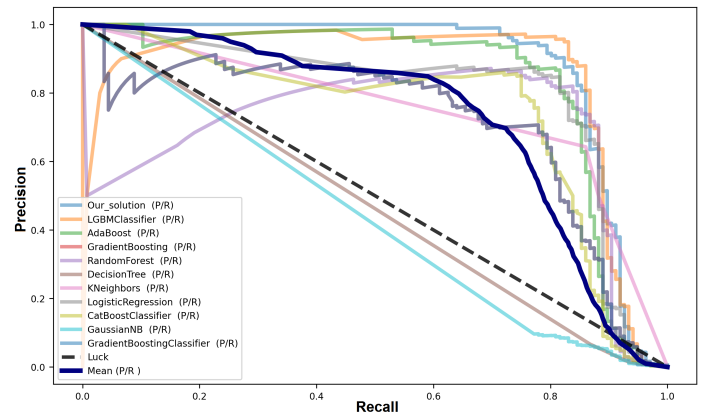


Fig. 6. Precision-Recall curve for the proposed solution

TABLE III
PERFORMANCE EVALUATION OF THE PROPOSED SOLUTION IN COMPARISON WITH VARIOUS MACHINE LEARNING ALGORITHMS

Classifier	Accuracy	Precision	Recall	F-measure	AUC
LGBM	0.9982	0.4782	0.8897	0.6221	0.9801
AD	0.9827	0.0792	0.9264	0.1460	0.9865
GB	0.9797	0.0692	0.9411	0.1289	0.9836
RF	0.9931	0.1772	0.9044	0.2963	0.9763
DT	0.9805	0.0669	0.8676	0.1242	0.9449
KNN	0.9990	0.6428	0.8602	0.7358	0.9407
LR	0.9731	0.0527	0.9338	0.0998	0.9803
CB	0.9794	0.0659	0.9044	0.1230	0.9773
NB	0.9757	0.0551	0.8823	0.1038	0.9687
Our Solution	0.9994	0.8068	0.8602	0.8327	0.9921

V. CONCLUSION

The huge number of transactions made every day in financial and banking institutions and the digitization of the business world made credit cards the first method of payment. Cardholders use their credit card as a wallet to pay for their purchase and transfer money. On the other side, a huge number of criminals try to attack those systems using various techniques to steal credit card information to do illegal transactions. So the necessity for stopping those attacks is needed and the enhancement of the existing methods is important today. Various methods are proposed by engineers and researchers as a solution for this issue based on machine learning technologies that are still not efficient. This work aims to propose a solution by combining the strong ability of three methods and exploiting the power of evolutionary algorithms for hyperparameters optimization. In addition, two resampling methods are combined as a solution for the imbalance issue in our credit card dataset. The synthetic minority oversampling and the edit nearest neighbor techniques. To illustrate the capability of the suggested solution, different experiments are conducted comparing this solution with various machine learning algorithms based on a publically available credit card dataset in terms of accuracy, precision, recall, f-measure, and AUC. In general, the out-coming results show the strong ability of the proposed methods in detecting fraud transactions. In terms of the precision score and AUC. For future work, the enhancements of this solution are proposed by adding different hyperparameters tuning and standing the behavior of the performance in terms of those hyperparameters added.

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